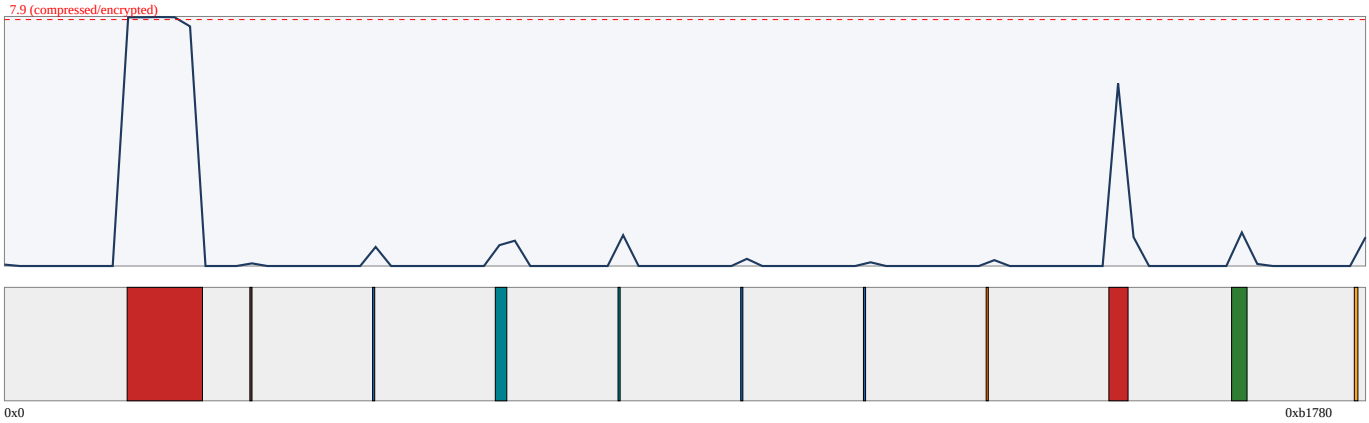
TV Intelligence Scanner — Firmware Analysis Report

Generated 2026-11-08T22:07:06+00:00 | v1.0.0 | analysis run #1 (completed, 2026-11-08T22:07:05Z to 2026-11-08T22:07:05Z)

1. File and integrity

File	orig_dump.bin
Size	726,912 bytes
SHA-256	`cce88a2118fc3fd1edd6b3fbd0cd38030335a5f964dd75e8d27ea476bf1089c7`
Library copy	/tmp/tmp2qb6piv8/data/TV_Firmware_Library/Acme/TESTTV55/Original/orig_dump.bin
Category / source	Original / local import
Manufacturer / model / board	Acme / TESTTV55 / ?
Original unchanged	YES — SHA-256 identical before and after analysis
Analysed copy	/tmp/tmp2qb6piv8/data/storage/workspace/fw_1/orig_dump.bin

2. Structure and entropy



Upper plot: Shannon entropy per block (0–8 bits/byte). Lower bar: top-level components (green filesystem, red kernel, purple bootloader, blue compressed, teal archive, orange executable, grey other).

Overall entropy 1.1147 bits/byte, chi-square 150266759.0, block size 8,192 B.

3. Metadata (each value with its source)

Attribute	Value	Source	Conf.
android_version	9	4 sources: extracted file 0x00030000_gzip/cpio_contents/system/build.prop; extracted file 0x00050000_zip/system/build.prop; extracted file 0x00040000_tar/system/build.prop; …	97%
board_platform	m7642	4 sources: property string @0x40635; extracted file 0x00030000_gzip/cpio_contents/system/build.prop; extracted file 0x00050000_zip/system/build.prop; …	97%
bootloader	U-Boot 2015.04-tv	string @0x14ee0	85%
build_date	Mon Mar 2 10:11:12 CST 2020	string @0x14ec4	60%
compression	bzip2, gzip, xz	validated signatures / headers	90%
cpu_architecture	ARM (32-bit)	2 sources: uImage header @0x10000 (CRC-verified); ELF headers (1 of 1 binaries)	97%
encryption_indicator	none — high-entropy regions are explained by recognised formats	entropy analysis	60%
endianness	little	ELF headers (1 binaries)	85%
filesystems	android_sparse	validated signatures	90%

Attribute	Value	Source	Conf.
kernel_cmdline	console=ttyS0 androidboot.hardware=tv	Android boot header @0x90000	90%
kernel_version	4.9.118	2 sources: string @0x14e75: Linux version 4.9.118 (builder@tv) (gcc version 6.3.1 (Linar; uImage name 'Linux-4.9.118' @0x10000	96%
os_family	Android	5 sources: extracted file 0x00030000_gzip/cpio_contents/system/build.prop; extracted file 0x00050000_zip/system/build.prop; extracted file 0x00040000_tar/system/build.prop; …	97%
os_family	Linux	kernel version string	85%
os_name	Linux	3 sources: extracted file 0x00030000_gzip/cpio_contents/etc/os-release; extracted file 0x00050000_zip/etc/os-release; extracted file 0x00040000_tar/etc/os-release	97%
os_version_id	4.9	3 sources: extracted file 0x00030000_gzip/cpio_contents/etc/os-release; extracted file 0x00050000_zip/etc/os-release; extracted file 0x00040000_tar/etc/os-release	97%
partition_info	androidboot.hardware=tv	string @0x9004e	70%
product_model	TESTTV55	4 sources: property string @0x4061b; extracted file 0x00030000_gzip/cpio_contents/system/build.prop; extracted file 0x00050000_zip/system/build.prop; …	97%
signing	Android Verified Boot, SHA256_RSA2048	vbmeta @0xb0000	90%
word_size	32-bit	ELF headers (1 binaries)	80%

4. Embedded components

Offset	Size	Type	Validated	Description
0x10000	40,237	uimage	yes	U-Boot uImage 'Linux-4.9.118': Linux kernel, ARM, compression gzip, data CRC OK
0x10040	40,173	gzip	yes	gzip stream: 40,173 B -> 40,140 B
0x20000	72	dtb	yes	Device tree blob (DTB), 72 B, version 17
0x30000	380	gzip	yes	gzip stream: 380 B -> 1,112 B
0x40000	6,144	tar	yes	POSIX tar archive, 5 entries
0x50000	778	zip	yes	ZIP archive, 5 entries (first: etc/os-release)
0x60000	140	xz	yes	xz stream: 140 B -> 24,000 B
0x70000	71	bzip2	yes	bzip2 stream: 71 B -> 30,000 B
0x80000	84	elf	yes	ELF 32-bit little-endian ARM executable
0x90000	10,240	android_boot	yes	Android boot image v0: kernel 5000 B, ramdisk 90 B
0x92000	90	gzip	yes	gzip stream: 90 B -> 252 B
0xa0000	8,256	android_sparse	yes	Android sparse image: 4 blocks of 4096 B (16,384 B expanded), 3 chunks
0xb0000	1,920	vbmeta	yes	Android Verified Boot metadata (vbmeta) v1.0, algorithm SHA256_RSA2048

Candidates rejected by validation: {'zip': 4}. Binwalk: {'skipped': 'disabled'}

5. Extraction (workspace only)

Offset	Type	Status	Method	Files	Note
0x10000	uimage	extracted	uImage payload	2	
0x30000	gzip	extracted	cpio (native)	6	1 link(s) not created.
0x40000	tar	extracted	tarfile	5	
0x50000	zip	extracted	zipfile	5	

Offset	Type	Status	Method	Files	Note
0x60000	xz	extracted	xz (native)	1	
0x70000	bzip2	extracted	bzip2 (native)	1	
0x92000	gzip	extracted	cpio (native)	2	
0xa0000	android_sparse	extracted	android sparse (native)	1	

Sensitive material present (contents not reproduced): 0x00030000_gzip/cpio_contents/etc/shadow, 0x00050000_zip/etc/shadow, 0x00040000_tar/etc/shadow

Configuration files (12): 0x00030000_gzip/cpio_contents/etc/os-release, 0x00030000_gzip/cpio_contents/etc/app.conf, 0x00030000_gzip/cpio_contents/system/build.prop, 0x00030000_gzip/cpio_contents/vendor/config/panel/panel_lc550ege.ini, 0x00050000_zip/etc/os-release, 0x00050000_zip/etc/app.conf, 0x00050000_zip/system/build.prop, 0x00050000_zip/vendor/config/panel/panel_lc550ege.ini, 0x00040000_tar/etc/os-release, 0x00040000_tar/etc/app.conf, 0x00040000_tar/system/build.prop, 0x00040000_tar/vendor/config/panel/panel_lc550ege.ini

6. Strings of interest

145 printable strings (6+ chars). Full list: /tmp/tmp2qb6piv8/data/storage/workspace/fw_1/run_1/strings.txt

Category	Offset	Value
kernel_version	0x14e75	4.9.118
uboot_version	0x14ee0	2015.04-tv
android_property	0x40600	ro.build.version.release 9
android_property	0x4061b	ro.product.model TESTTV55
android_property	0x40635	ro.board.platform m7642
build_date	0x14ec4	Mon Mar 2 10:11:12 CST 2020
partition_name	0x9004e	androidboot.hardware=tv

Notes

Findings describe the file's structure; they do not establish that the firmware is authentic or compatible with any TV. Encryption statements are indicators based on entropy and the absence of recognised formats. Version values carry the source they were read from; file-name based values are the weakest evidence.